

Example

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1 Quarto Example

This is an example of a Quarto document, you will need to install Quarto from <https://quarto.org/>. This package comes with several common R packages, it is also recommended to install:

```
install.packages(c('gt', # general table formatting
  'gtsummary', # for univariate and regression tables
  'here', # where you are working (better when working with git)
  'knitr', # r package for working with documents
  'quarto', # R package for working with quarto
  'rmarkdown', # creating documents
  'tidymodels', # modern interface for fitting models
  'tidyverse', # modern R language
  'tinytex', # needed for pdf documents
  'devtools' # package creation development tools
))

remotes::install_github("Pakillo/grateful") # for package citation
```

To create pdf documents it is also recommended to run

```
tinytex::install_tinytex()
```

Table 1: first 5 records and 5 columns of the data

subject_id	group	category	measure_1	measure_2
SYN-001	A	X	4.67	5.71
SYN-002	A	Y	5.43	4.76
SYN-003	A	Y	6.00	5.20
SYN-004	A	Y	4.74	6.21
SYN-005	B	Y	5.07	6.50

2 Table One

3 Cross Reference

You can cross-reference tables, listings, figures and page numbers

You can use the `labels` you created to reference the tables like Table 2 or items in the `bib.bib` file such as Al-Helali et al. (2021).

You can even reference later items such as Figure 2 or sections like Section 5 or even the page number that something happens on like page 6.

4 Pairwise t-tests

Suppose we wanted to visualize all pairwise t-tests (we will only do first four but you get the idea). There are other ways to do this but this is a good enough example of repeated analysis.

Table 2: An example of a Table 1

Characteristic	A N = 27 ¹	B N = 33 ¹	p-value	Overall ¹
category			0.94 ²	
X	12 (44%)	13 (39%)		25 (42%)
Y	10 (37%)	14 (42%)		24 (40%)
Z	5 (19%)	6 (18%)		11 (18%)
measure_1			0.67 ³	
Mean (SD)	5.34 (0.71)	5.43 (0.80)		5.39 (0.76)
Median (Q1, Q3)	5.43 (4.67, 5.99)	5.40 (5.07, 5.87)		5.40 (4.77, 5.92)
Median (Min - Max)	5.43 (4.32 - 6.71)	5.40 (3.88 - 7.66)		5.40 (3.88 - 7.66)
measure_2			0.62 ³	
Mean (SD)	5.12 (0.71)	5.19 (0.80)		5.16 (0.76)
Median (Q1, Q3)	4.99 (4.72, 5.73)	5.17 (4.77, 5.70)		5.14 (4.74, 5.72)
Median (Min - Max)	4.99 (3.76 - 6.36)	5.17 (3.63 - 6.72)		5.14 (3.63 - 6.72)
measure_3			0.27 ³	
Mean (SD)	3.99 (1.63)	3.60 (2.00)		3.77 (1.83)
Median (Q1, Q3)	3.72 (2.88, 5.19)	3.34 (1.83, 4.68)		3.51 (2.52, 4.92)
Median (Min - Max)	3.72 (1.39 - 7.66)	3.34 (0.76 - 8.41)		3.51 (0.76 - 8.41)
measure_4			0.29 ³	
Mean (SD)	5.92 (2.12)	5.35 (2.25)		5.61 (2.19)
Median (Q1, Q3)	5.45 (4.30, 7.49)	4.95 (3.70, 6.94)		5.31 (3.90, 7.10)
Median (Min - Max)	5.45 (2.83 - 10.69)	4.95 (1.68 - 10.40)		5.31 (1.68 - 10.69)
measure_5			0.29 ³	
Mean (SD)	7.2 (3.7)	6.2 (2.7)		6.7 (3.2)
Median (Q1, Q3)	6.4 (4.1, 8.7)	6.0 (4.9, 7.6)		6.1 (4.6, 7.9)
Median (Min - Max)	6.4 (1.3 - 15.8)	6.0 (2.0 - 12.9)		6.1 (1.3 - 15.8)
measure_6			0.68 ³	
Mean (SD)	3.99 (1.89)	4.38 (2.22)		4.21 (2.07)
Median (Q1, Q3)	3.65 (2.36, 5.19)	4.10 (2.59, 5.43)		3.78 (2.54, 5.42)
Median (Min - Max)	3.65 (0.81 - 8.23)	4.10 (1.80 - 10.83)		3.78 (0.81 - 10.83)

¹n (%)²Fisher's exact test³Wilcoxon rank sum exact test

4.1 between measure_1 and measure_2

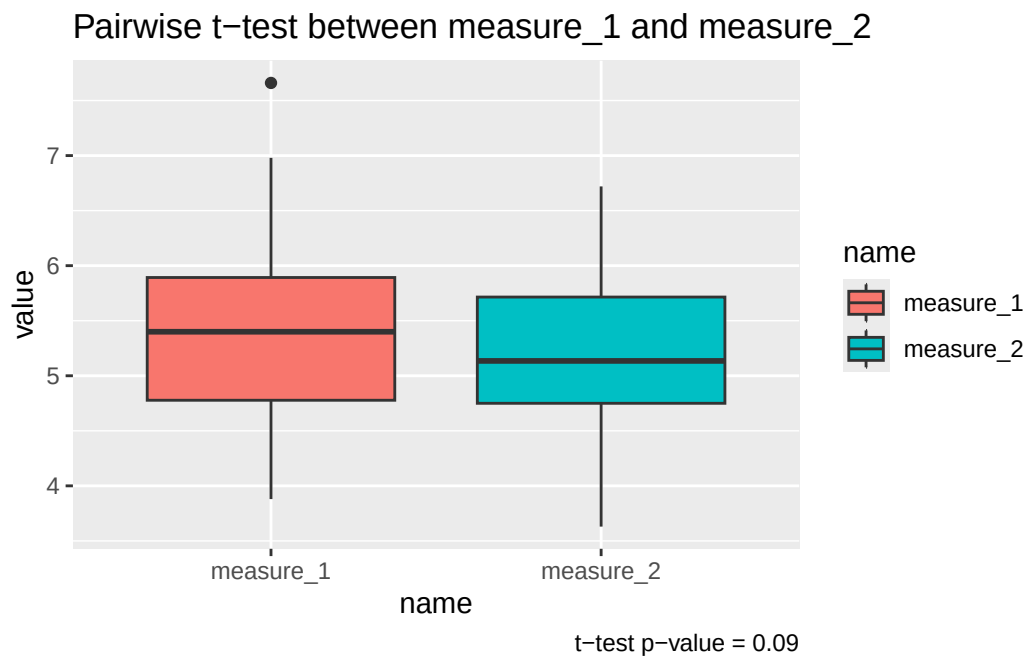


Figure 1: Pairwise t-test between measure_1 and measure_2

4.2 between measure_1 and measure_3

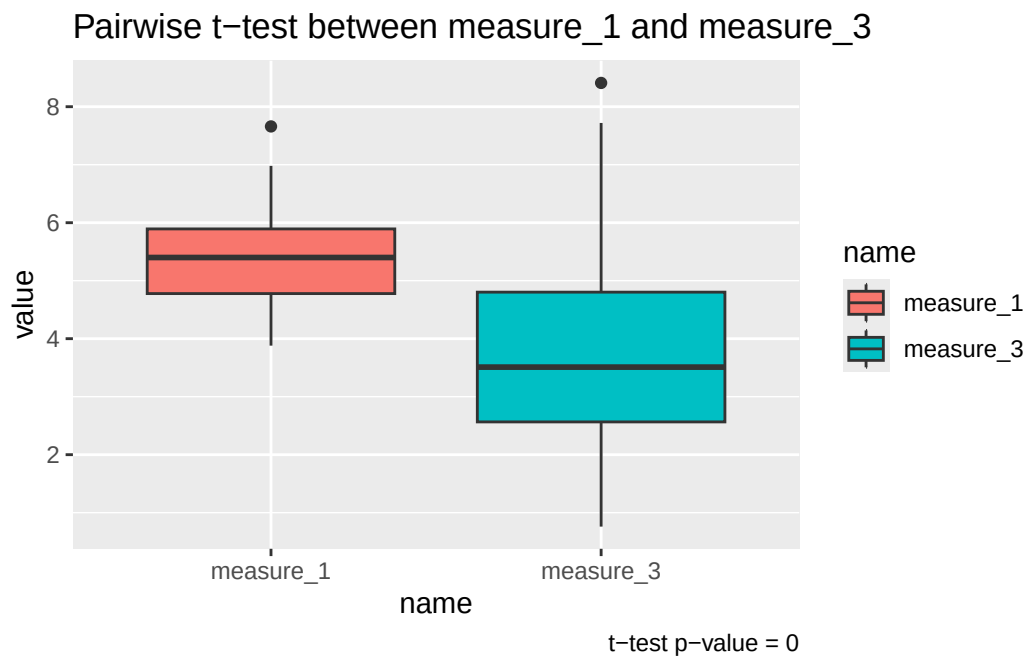


Figure 2: Pairwise t-test between measure_1 and measure_3

4.3 between measure_1 and measure_4

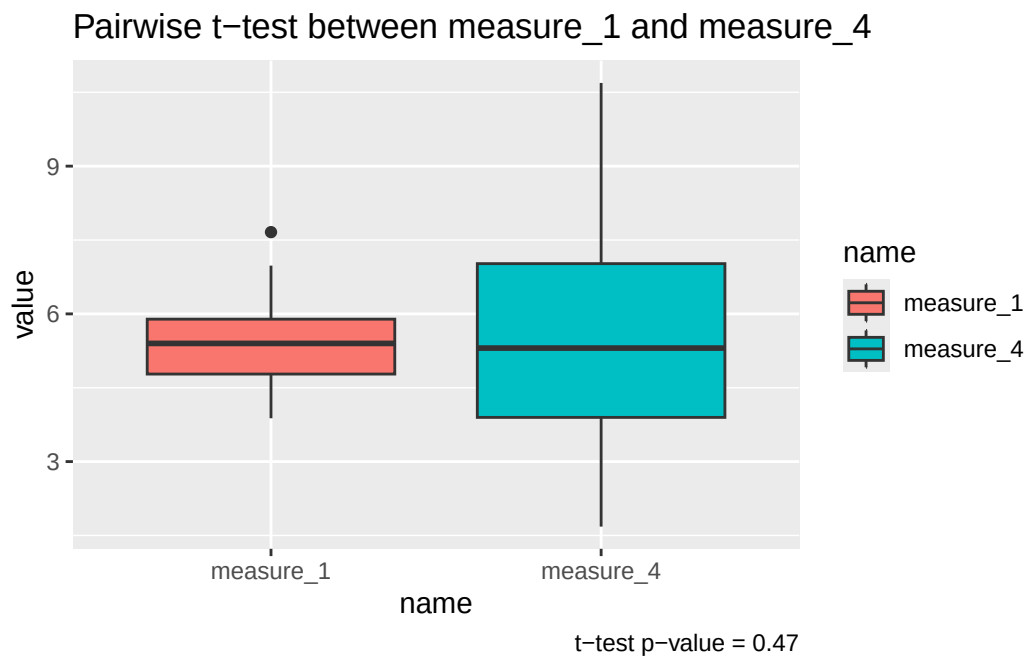


Figure 3: Pairwise t-test between measure_1 and measure_4

4.4 between measure_1 and measure_5

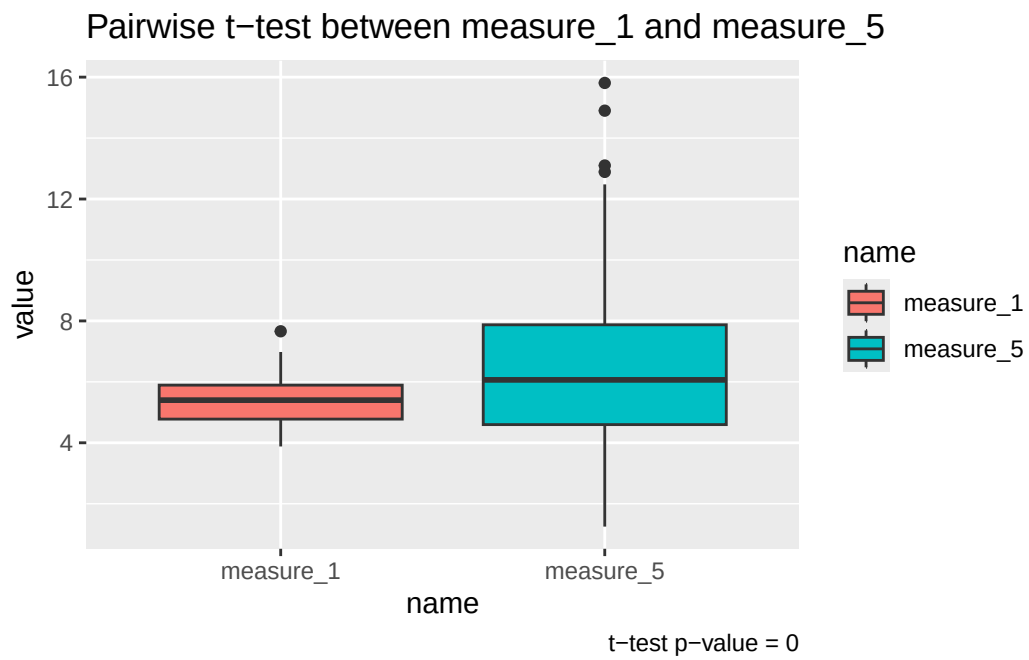


Figure 4: Pairwise t-test between measure_1 and measure_5

5 Software

This work was completed using R v. 4.6.1 (R Core Team 2026) and the following R packages: bookdown v. 0.48 (Xie 2016, 2026a), grateful v. 0.3.0 (Rodriguez-Sanchez and Jackson 2025), gt v. 1.3.0 (Iannone et al. 2026), gtsummary v. 2.6.1 (Sjoberg et al. 2021), here v. 1.0.2 (Müller 2025), knitr v. 1.52 (Xie 2014, 2015, 2026b), rmarkdown v. 2.32 (Xie et al. 2018, 2020; Allaire et al. 2026), tidyverse v. 2.0.0 (Wickham et al. 2019).

References

- Al-Helali, Baligh, Qi Chen, Bing Xue, and Mengjie Zhang. 2021. “A New Imputation Method Based on Genetic Programming and Weighted KNN for Symbolic Regression with Incomplete Data.” *Soft Comput.* 25 (8): 5993–6012.
- Allaire, JJ, Yihui Xie, Christophe Dervieux, et al. 2026. *rmarkdown: Dynamic Documents for r*. <https://github.com/rstudio/rmarkdown>.
- Iannone, Richard, Joe Cheng, Barret Schloerke, et al. 2026. *gt: Easily Create Presentation-Ready Display Tables*. <https://gt.rstudio.com>.
- Müller, Kirill. 2025. *here: A Simpler Way to Find Your Files*. <https://doi.org/10.32614/CRAN.package.here>.
- R Core Team. 2026. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://doi.org/10.32614/R.manuals>.
- Rodriguez-Sanchez, Francisco, and Connor P. Jackson. 2025. *grateful: Facilitate Citation of R Packages*. <https://pakillo.github.io/grateful/>.
- Sjoberg, Daniel D., Karissa Whiting, Michael Curry, Jessica A. Lavery, and Joseph Larmarange. 2021. “Reproducible Summary Tables with the Gtsummary Package.” *The R Journal* 13: 570–80. <https://doi.org/10.32614/RJ-2021-053>.
- Wickham, Hadley, Mara Averick, Jennifer Bryan, et al. 2019. “Welcome to the tidyverse.” *Journal of Open Source Software* 4 (43): 1686. <https://doi.org/10.21105/joss.01686>.
- Xie, Yihui. 2014. “knitr: A Comprehensive Tool for Reproducible Research in R.” In *Implementing Reproducible Computational Research*, edited by Victoria Stodden, Friedrich Leisch, and Roger D. Peng. Chapman; Hall/CRC.
- Xie, Yihui. 2015. *Dynamic Documents with R and Knitr*. 2nd ed. Chapman; Hall/CRC. <https://yihui.org/knitr/>.
- Xie, Yihui. 2016. *bookdown: Authoring Books and Technical Documents with R Markdown*. Chapman; Hall/CRC. <https://pkg.yihui.org/bookdown/>.
- Xie, Yihui. 2026a. *bookdown: Authoring Books and Technical Documents with r Markdown*. <https://github.com/rstudio/bookdown>.
- Xie, Yihui. 2026b. *knitr: A General-Purpose Package for Dynamic Report Generation in R*. <https://yihui.org/knitr/>.
- Xie, Yihui, J. J. Allaire, and Garrett Golemund. 2018. *R Markdown: The Definitive Guide*. Chapman; Hall/CRC. <https://pkg.yihui.org/rmarkdown-book/>.

Xie, Yihui, Christophe Dervieux, and Emily Riederer. 2020. *R Markdown Cookbook*. Chapman; Hall/CRC.
<https://pkg.yihui.org/rmarkdown-cookbook/>.